A conditional Lean formalization of an 88.45% critical-line proportion for ξ'

An explicit two-component mollifier

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Abstract

We describe a Lean derivation of a fixed asymptotic proportion strictly greater than 88.45% for the zeros of the derivative of the Riemann xi-function on the critical line, counted with analytic multiplicity. The result has two explicit hypotheses: a uniform incomplete inverse-phase estimate and a four-length Deshouillers–Iwaniec estimate. We give their full parameter ranges, coefficient restrictions and quantifier order. Both estimates are established in the literature: the incomplete estimate follows by completion from the composite-modulus Kloosterman bound, and the second is Deshouillers–Iwaniec's Lemma 1. *Conditional* refers to the formalization boundary: their proof terms are not supplied in Lean here, not to new conjectural assumptions. From these inputs the formal chain derives the shifted moment for a height-dependent seven-vector coefficient family, including all 49 ordered pairs and the actual approximate functional equation remainders. It then connects the density-oriented arithmetic main term to a fixed two-component mollifier, a kernel-checked rational gate, and the actual xi-prime counting functions. A separate Python interval calculation supplies a sharper numerical check but is not trusted by Lean. We distinguish formal deduction, correspondence with the cited analytic inputs, and independent mathematical review. No simple-zero, optimality or publication-priority claim is made.

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1 The statement and its trust boundary

Write

$$\xi(s) = H(s)\zeta(s), \qquad H(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2).$$

Let $\mathcal{N}_{\xi'}(X)$ be the total xi-prime zero count in the positive-height window $(0, X]$, and $\mathcal{N}_{\xi',c}(X)$ its critical-line subcount. Both use analytic multiplicity. More precisely, these are the counts `Zeta23.XiPrime.Ncount 0 X` and `Zeta23.XiPrime.N0 0 X` in the pinned Alpöge–Furman (AF) library; they are not its simple-zero counts. The formal counting normalization is used throughout this manuscript, rather than replacing the denominator by the zeta zero count.

Theorem 1 (Conditional critical-line proportion)**.** *Assume Hypotheses 1 and 2 below. There exist real numbers $b > 0.8845$ and X_0 such that, for every real $X \geq X_0$,*

$$b\,\mathcal{N}_{\xi'}(X) \leq \mathcal{N}_{\xi',c}(X). \tag{1}$$

In particular, the denominator is eventually positive and $\mathcal{N}_{\xi',c}(X)/\mathcal{N}_{\xi'}(X) > 0.8845$ for all sufficiently large real X. Consequently the asymptotic lower proportion is strictly greater than 0.8845, with a fixed positive margin.

The first assertion is `XiPrime8845.WuProportion.strict_proportion`; the quotient assertion is `WuProportion.ratio_eventually_gt`. The proof obtains a fixed margin before allowing the terminal height to grow. It does not merely prove a strict inequality whose margin could tend to zero.

The two hypotheses are the propositions `PRZZKloostermanInputs.WeilIncomplete` and `PRZZKloostermanInputs.DeshouillersIwaniecLemma1`. These are established analytic estimates, not conjectures about zeros. Hypothesis 1 follows from the composite-modulus Kloosterman bound by the completion argument in Section 2 [3, Corollary 11.12]; Hypothesis 2 is the four-length estimate of [4, Lemma 1]. The word *conditional* in the title therefore describes the formal proof boundary: their proofs are not formalized in this development. Independent review of the precise source-to-Lean specification remains important. No proof term for either proposition is supplied here. In particular, the name Weil does not suppress the completion, prime-power and composite-modulus obligations specified below. The former public PRZZ witness axiom has instead been replaced by a theorem deduced from these two arguments. The mathematical implication proved by Lean and the validity of the external analytic specifications are separate matters.

Historically, the mollified-moment method goes back to Levinson [13]; Conrey studied critical-line zeros of derivatives of ξ [14]. The long-mollifier scale $\theta < 4/7$ and the use of DI estimates follow the line of Conrey [10] and the twisted mean-square work of Balasubramanian–Conrey–Heath-Brown [11]. The two-component coefficient construction also draws on Feng [9]; its unrestricted convolution is specified below. The construction uses Wu's argument-variation strategy [5], the long-mollifier moment architecture of Pratt–Robles–Zaharescu–Zeindler [6], and the pinned AF analytic/counting infrastructure [7, 8], implemented in Lean 4 [15] with Mathlib [16]. The following sections give a proof guide to the current source, not a claim that a short prose argument replaces every contour, integrability and boundary lemma in that source.

For context, Wu's one-component mollifier gives the unconditional on-line proportion 0.86957, without requiring the counted zeros to be simple [5, Section 3]. Wu uses the zeta zero count as the denominator; the Riemann–von Mangoldt asymptotics identify its leading term with the total ξ' zero count used here. AF [7, Remark 7.1 in the cited version] gives 0.85838, or 0.86864 with its quartic window, for zeros of ξ' that are *simple and on the line*, also unconditionally. Simplicity is a property of the zeros counted there, not an additional hypothesis. Under Hypotheses 1 and 2, our fixed two-component witness certifies a proportion strictly above 0.8845 with multiplicity, without a simplicity conclusion.

2 The two explicit exponential-sum hypotheses

Put $e(x) = \exp(2\pi i x)$. If $(n, q) = 1$, let $\bar{n}_q$ be the least nonnegative representative of the inverse of n modulo q. It is a modular inverse, never the real reciprocal. At modulus 1 the representative is 0, as in the Lean definition using $\mathbb{Z}/q\mathbb{Z}$. All occurrences below are guarded by the relevant coprimality conditions.

2.1 The incomplete Weil input

For $q \geq 1$, $a \in \mathbb{Z}$, and $M, H \in \mathbb{Z}_{\geq 0}$, set

$$S_q(a; M, H) = \sum_{\substack{M < n \leq M+H \\ (n,q)=1}} e\left(\frac{a\bar{n}_q}{q}\right).$$

Hypothesis 1 (Uniform incomplete inverse-phase estimate)**.** For every real $\varepsilon > 0$ there exists $C_\varepsilon > 0$ such that, for every integer $q \geq 1$, every $a \in \mathbb{Z}$, and all nonnegative integers M, H,

$$|S_q(a; M, H)| \leq C_\varepsilon q^{1/2+\varepsilon} \gcd(|a|, q) \left(1 + \frac{H}{q}\right). \tag{2}$$

Thus C_ε is chosen *before* q, a, M, H. There is no assumption that $(a, q) = 1$, that q is prime, or that $H \leq q$. We use the convention $\gcd(0, q) = q$, so $a = 0$ is included. The gcd is to the first power in this precise contract; replacing it by 1 would change the hypothesis. The factor $1 + H/q$

retains long intervals. The case $H = 0$ is the empty sum. This is an unweighted input; smooth weights and their norms are supplied later.

For comparison, the separate definition `WeilPrime` asks for

$$\left|\sum_{n=1}^{p-1} e\Big(\frac{an + b\bar{n}_p}{p}\Big)\right| \le 2\sqrt{p} \quad (p > 2 \text{ prime},\ (a,p) = (b,p) = 1).$$

This is the prime-field statement associated with Weil [1, 2]. It is not the hypothesis of Theorem 1. Deriving (2) from that statement requires estimates at prime powers, Chinese-remainder assembly and completion of intervals, including the logarithmic/divisor losses absorbed by q^{ε}. The repository deliberately does not assert a proved `WeilPrime` ⇒ `WeilIncomplete` implication. PRZZ invokes an incomplete estimate in its Type I argument [6, p. 21]; Hypothesis 1 records the exact completion specification consumed here.

For an explicit classical route, use the composite-modulus estimate [3, Corollary 11.12, equation (11.16)]:

$$|S(h,a;q)| \le \tau(q) q^{1/2} \gcd(h,a,q)^{1/2}.$$

Put $g = \gcd(|a|, q)$. For an interval J of fewer than q consecutive integers, finite Fourier inversion gives

$$\sum_{\substack{n\in J\\(n,q)=1}} e(a\bar{n}_q/q) = \frac{1}{q}\sum_{h \bmod q}\left(\sum_{n\in J} e(-hn/q)\right) S(h,a;q).$$

The normalized sum of absolute values of the interval Fourier coefficients is $O(\log(2q))$, so this is $O(\tau(q)\sqrt{qg}\log(2q))$. Each complete period is the Ramanujan sum $c_q(a)$. By von Sterneck's formula $c_q(a) = \mu(q/g)\varphi(q)/\varphi(q/g)$ and $\varphi(q) \le g\varphi(q/g)$, its modulus is at most g. Splitting H into complete periods and one such interval therefore gives

$$|S_q(a;M,H)| \ll_{\varepsilon} q^{1/2+\varepsilon} g^{1/2}(1 + H/q).$$

Here $g \le \sqrt{qg}$ and the divisor/logarithmic loss is absorbed into q^{ε}; $q = 1$ is immediate. This stronger estimate implies (2). Thus the first-power gcd is a permissible weakening, not a necessary cost of the zero mode. This completion argument is supplied in prose, not as a Lean proof term for Hypothesis 1.

2.2 The Deshouillers–Iwaniec input

Let $C, D, U, V \ge 1$ be real numbers, $\rho \ge 1$ an integer, and $b : \mathbb{N}\times\mathbb{N} \to \mathbb{C}$. Define

$$\mathscr{S}(C,D,U,V;\rho,b) = \sum_{1\le c\le \lfloor C\rfloor}\ \sum_{\substack{1\le d\le \lfloor D\rfloor\\(c,\rho d)=1}} \left|\sum_{1\le u\le \lfloor U\rfloor}\ \sum_{\substack{1\le v\le \lfloor V\rfloor\\(v,c)=1}} b(u,v) e\Big(\frac{u\,\overline{\rho v d}_c}{c}\Big)\right|. \tag{3}$$

In particular, the absolute value is *inside* the two outer sums. The outer and inner coprimality restrictions guarantee that $\rho v d$ is invertible modulo c. There is no coprimality restriction on u. Set

$$\mathscr{B}_{\varepsilon}(C,D,U,V;\rho) = (CDUV)^{1/2+\varepsilon}\{(CD)^{1/2} + (U+V)^{1/4}\mathscr{R}^{1/4}\}, \tag{4}$$
$$\mathscr{R} = CD(U+\rho V)(C+\rho V^2) + \rho UV^2D^2.$$

Hypothesis 2 (Uniform DI four-length estimate)**.** For every real $\varepsilon > 0$ there exists $K_{\varepsilon} > 0$ such that, for every $C, D, U, V \ge 1$, every integer $\rho \ge 1$, and every complex array b satisfying

$$|b(u,v)| \le 1 \quad (1 \le u \le \lfloor U\rfloor,\ 1 \le v \le \lfloor V\rfloor),$$

one has

$$\mathscr{S}(C,D,U,V;\rho,b) \le K_{\varepsilon}\,\mathscr{B}_{\varepsilon}(C,D,U,V;\rho). \tag{5}$$

The unit bound holds on the full rectangular coefficient set, including indices not used by a particular coprimality filter. No smoothness or factorization of $b(u,v)$ is assumed. One array is used across all (c,d) in (3); an arbitrary array depending on c,d would be a different input. Most importantly, the quantifiers are

$$\forall \varepsilon > 0\ \exists K_\varepsilon > 0\ \forall C,D,U,V \geq 1\ \forall \rho \geq 1\ \forall b.$$

The constant cannot depend on ρ, the four lengths, or the coefficient array. Real upper bounds are implemented by natural floors.

An auxiliary loss ρ^δ of arbitrarily small positive order can be absorbed, provided the estimate is otherwise uniform in ρ. Indeed, put $X = CDUV \geq 1$. Then $(U+V)\mathscr{R} \geq \rho^2 X$, so $\mathscr{B}_\varepsilon \geq \rho^{1/2} X^{3/4+\varepsilon}$. For $\rho \geq X^{1/2}$ this dominates the trivial bound $\mathscr{S} \leq X$. For $1 \leq \rho < X^{1/2}$, an otherwise uniform bound $\mathscr{S} \ll_\delta \rho^\delta \mathscr{B}_\delta$ gives $\rho^\delta \mathscr{B}_\delta \leq \mathscr{B}_\varepsilon$ on choosing $\delta = 2\varepsilon/3$. This observation removes such an epsilon-power loss, not arbitrary dependence of the implied constant on ρ, and does not replace the source correspondence check below.

The source specified for this contract is Deshouillers–Iwaniec, *Power mean-values for Dirichlet's polynomials and the Riemann zeta-function, II*, Lemma 1, p. 310 [4]. The publisher lists volume 43 (1984), pp. 305–312; the volume 48 printed in PRZZ's reference is not used here. The author compared the original scanned Lemma 1 with the displayed inverse phase, both coprimality filters and the four-length expression used here. The source writes its coefficient array as $c(u,v)$; here it is renamed $b(u,v)$ to distinguish it from the outer index c. The symbols $(C,D,U,V,\rho,b(u,v))$ agree with the code's argument names. Independent review is invited to check that comparison and the complete source-to-Lean translation, including uniformity and quantifier order. The author's source comparison is not a completed independent audit and is not an additional theorem supplied by the axiom audit.

2.3 How the inputs enter the proof

The definitions in `PRZZKloostermanInputs.lean` are propositions, not declarations of analytic axioms. Intermediate modules use a class `PRZZProvedInput.DeepInputs` with exactly two fields, one for each hypothesis. No default instance asserts those fields. The four headline theorems expose them as separate ordinary arguments.

Type I partial summation uses (2); the coupled Type II estimate uses (5). Cutoff seminorms, arithmetic coefficients, dyadic multiplicities, and the time/Mellin masses are not part of either input's hidden constant. They are retained and estimated in the adapters. This distinction is necessary for the height-dependent profiles below.

3 Fixed data and the height-dependent coefficient family

Fix

$$R = 1.04440033, \qquad \theta = \frac{4}{7} - 10^{-6}, \qquad L = \log T, \quad y = T^\theta, \quad a = \frac{1}{2} - \frac{R}{L}. \tag{6}$$

The operator and profiles are

$$\begin{aligned} q(x) &= 1 - 2.27043254x + 0.304358115x^2 + 0.16793368x^3 \\ &\quad + 4.34152016x^4 - 5.3105844x^5 + 1.7701948x^6, \end{aligned} \tag{7}$$

$$P_1(x) = -0.30528514x + 2.50038954x^2 - 1.12177152x^3 - 0.07333288x^4, \tag{8}$$

$$P_2(x) = 1.89779072x + 0.56594804x^2 + 0.07023055x^3. \tag{9}$$

Every finite decimal here is an exact rational number. In particular $q(0) = P_1(1) = 1$ and $P_1(0) = P_2(0) = 0$. For $1 \leq n \leq y$ put $x_n = \log(y/n)/\log y$ and

$$a_n(T) = \mu(n) P_1(x_n) + \frac{(\mu * \Lambda * \Lambda)(n)}{(\log y)^2} P_2(x_n); \tag{10}$$

put $a_n(T) = 0$ outside this range. Here $*$ is Dirichlet convolution; the second component is of Feng type [9]. There is no squarefree restriction and no extra 2!. Define

$$\Psi_T(s) = \sum_{1\le n\le \lfloor y\rfloor} a_n(T) n^{-s-R/L}, \qquad U_T(s) = q(-L^{-1}\partial_s)\zeta(s). \tag{11}$$

Thus $\Psi_T(a+\mathrm{i}t) = \sum a_n(T) n^{-1/2-\mathrm{i}t}$. The differential operator acts on zeta, not on the mollifier coefficients.

The elementary estimates $|\mu(n)| \le 1$ and $|(\mu * \Lambda * \Lambda)(n)| \le (\log n)^2$ give $|a_n(T)| \le 6.53474839$ for $T \ge 2$. This coefficient bound alone does not assert a moment theorem for arbitrary bounded coefficient arrays. The relevant structured family has seven basis vectors:

$$3\mu(n)x_n^k \quad (1\le k\le 4), \qquad 3\,\frac{(\mu*\Lambda*\Lambda)(n)}{(\log y)^2}x_n^k \quad (1\le k\le 3),$$

all with the same original support indicator. Denote them by $b_i(\theta, T; n)$, $0 \le i < 7$. The witness coordinates z_i are the seven coefficients in P_1, P_2, divided by 3; hence $|z_i| \le 1$. The identity $a_n(T) = \sum_i z_i b_i(\theta, T; n)$ is exact at every height. Both the logarithmic basis and its support depend on T. The proved envelopes are $|b_i| \le 6$ and $|\sum_i z_i b_i| \le 84$ for the larger box $|z_i| \le 2$.

4 Uniform moments from the two inputs

For coefficient sequences c, d, write $D_c(s) = \sum_{1\le n\le N} c_n n^{-s}$ and define

$$\begin{aligned}\mathcal{I}_T(c,d;A,B) = \int_T^{2T} &\zeta(\tfrac12+\alpha+\mathrm{i}t)\zeta(\tfrac12+\beta-\mathrm{i}t)\\ &\cdot D_c(\tfrac12+\mathrm{i}t)\overline{D_d(\tfrac12+\mathrm{i}t)}\Phi(t/T)\,\mathrm{d}t,\end{aligned} \tag{12}$$

where $\alpha = A/\log T$, $\beta = B/\log T$ and Φ is a fixed real C^∞ function with support in $[1,2]$. Positivity is not required for this mixed-moment statement. It is required when positive moments are used for counting.

For positive integers m, n, put $g = (m,n)$ and $\ell = [m,n]$ and set

$$\begin{aligned}\mathcal{K}_T(m,n;A,B) = \frac{1}{\ell}\,\frac{g^{\alpha+\beta}}{m^\beta n^\alpha}\int_T^{2T} \Big\{&\zeta(1+\alpha+\beta)\\ &+\zeta(1-\alpha-\beta)\left(\frac{2\pi mn}{tg^2}\right)^{\alpha+\beta}\Big\}\Phi(t/T)\,\mathrm{d}t,\end{aligned} \tag{13}$$

$$\mathcal{G}_T(c,d;A,B) = \sum_{m,n=1}^{N} c_m\overline{d_n}\mathcal{K}_T(m,n;A,B).$$

Positive real bases in complex powers use the real logarithm. These formulas are used only with $A + B \ne 0$ and sufficiently large T. At $A + B = 0$ the raw bracket in Lean is formally `2 riemannZeta 1`, not the removable holomorphic extension. No extension across that divisor is used in this proof.

Proposition 2 (Uniform finite-family moment)**.** *Assume Hypotheses 1 and 2. Fix Φ as above, $\tau > 0$, $S \ge 0$ and $r > 0$. There exist $C > 0$ and T_0 such that, for every $T \ge T_0$, every $\theta > 0$, every integer $N \ge 1$, all vectors $z, w \in \mathbb{C}^7$ with $|z_i|, |w_i| \le 2$, and all $A, B \in \mathbb{C}$ satisfying*

$$N + 1 \le T, \quad N \le T^{4/7}, \quad \log(T^\theta) \ge 1, \quad N \le T^\theta, \qquad |A|, |B| \le S, \quad r \le |A+B|,$$

the sequences $c = \sum_i z_i b_i(\theta, T)$ and $d = \sum_i w_i b_i(\theta, T)$ satisfy

$$|\mathcal{I}_T(c,d;A,B) - \mathcal{G}_T(c,d;A,B)| \le CT^\tau\big(N^{7/4} + T^{1/2}N^{7/8}\big). \tag{14}$$

This is the specialization of `PRZZFamilyMomentOptimal.moment_bound` obtained by fixing a smooth monotone plateau seed and $K = 4$. The seed equals 1 on $(-\infty, 1/2]$, equals 0 on $[1, \infty)$, and takes values in $[0, 1]$. The witness bridge constructs it from a smooth transition function. The constant and height threshold precede θ, N, z, w, A, B. They may depend on the fixed seed, Φ, τ, S, r and the input constants. This is not a uniform assertion over arbitrary time weights or arbitrary bounded coefficient sequences.

4.1 Uniform smooth cutoffs and Mellin mass

The support scale is not frozen when a profile is inserted into a dyadic arithmetic cutoff. A useful intermediate class consists of smooth $f : (0, \infty) \to \mathbb{C}$, supported in $[a, b]$ with $0 < a \le b$, $\log(b/a) \le w$, and

$$|x^n f^{(n)}(x)| \le U_n \quad (n \ge 0).$$

For $\widehat{f}_M(s) = \int_0^\infty f(x) x^{s-1}\,\mathrm{d}x$, the implemented class estimate is

$$\int_{\mathbb{R}} (1 + |v|)|\widehat{f}_M(\mathrm{i}v)|\,\mathrm{d}v \le 4w(U_0 + U_1 + 3U_2 + U_3) \int_{\mathbb{R}} \frac{\mathrm{d}v}{1 + v^2}. \tag{15}$$

It follows from the logarithmic-coordinate derivative estimates and integration by parts, with endpoint terms removed by smooth compact support. The constants precede the function and the support scale. `PRZZProfileScale` supplies the dimensionless jets for the actual height-dependent profiles, and `PRZZProfileMellin.member_mass` supplies (15). These bounds are carried through the Type I/II and Heath–Brown block estimates [12] before the outer row sums are taken. A fixed-function bound alone would not justify Proposition 2.

4.2 What is assembled before the final bound

The proof uses the actual Gamma-ratio AFE, not a replacement of its kernel by definition. The moment error is split into the AFE replacement error, the nonzero Poisson field, and the Gamma main-term defect. The nonzero field is further split into its paired central body and its body tail. The source cutoff, row Fourier cutoff and their transfer bands are kept distinct until their correction terms are bounded. Zero-frequency mismatch, spatial complements, boundary corrections, freezing errors and large-scale terms are included before the cutoffs are removed. Positive and negative frequencies and the two dual branches are retained.

At the final assembly point the algebra has the form

$$E = (E - \mathcal{R} - \mathcal{G}_\Gamma) + (\mathcal{R} - \mathcal{B}) + \mathcal{B} + \mathcal{G}_\Gamma, \qquad E = \mathcal{I}_T - \mathcal{G}_T.$$

The four suppliers used by `PRZZFamilyMomentOptimal.basis_bound` are respectively `PRZZCappedMainTerm.zero_removed_reduction`, `PRZZCappedBodyTail.nonzero_reduction`, `PRZZBasisPairedBody.bound`, and `PRZZCappedGammaDefect.mixed_absorbed`. Here $\mathcal{B}$ is the actual paired central body, not an independent proxy. The Weil and DI hypotheses enter its estimate. The preceding reductions also include the Gamma-to-density kernel comparison `PRZZCappedMainTerm.mainterm_absorbed`.

For each of the 49 ordered basis pairs this gives a common upper bound U. Sesquilinearity then gives a bound

$$\sum_{i,j} |z_i|\,|w_j|\,U \le 196U.$$

The coordinate vectors may vary with height. Polarization is available as an algebraic bridge, but it does not replace the analytic uniformity of the 49 basis estimates.

4.3 Witness specialization and loss margins

`PRZZWitnessBridge.witness_estimate` supplies $N = \lfloor T^\theta \rfloor$ for each fixed $0 < \theta < 4/7$, and identifies the compact-window integral with the public all-real-time integral, since $\Phi(t/T)$ vanishes outside $[T, 2T]$. Thus `PRZZ.przz41_case3_K2_witness` is now a theorem with explicit Weil/DI arguments, not a public analytic axiom.

Every $\tau > 0$ is available. For the value of θ in (6), choosing $\tau = \frac{1}{4\times 10^6}$ gives

$$T^{\tau-1}(y^{7/4} + T^{1/2}y^{7/8}) = T^{-\frac{3}{2\,000\,000}} + T^{-\frac{1}{1\,600\,000}} \longrightarrow 0.$$

The public counting route uses the also admissible loss $\frac{7}{16\,000\,000}$; its two normalized exponents are $-\frac{21}{16\,000\,000}$ and $-\frac{7}{16\,000\,000}$. These are distinct choices in two proved specializations. The smaller positive margin from the second term must not be replaced by the first term's margin.

4.4 Orientation of the shifts and the historical correction

In (12), let m, n be the coefficient indices (d, e in `PRZZInterface`) and k, k' the zeta-series indices. The diagonal $mk = nk'$ gives $k = (n/g)h$, $k' = (m/g)h$ with $g = (m, n)$, so $k^{-\alpha}(k')^{-\beta} = (n/g)^{-\alpha}(m/g)^{-\beta}h^{-\alpha-\beta}$. Thus (13) places β on m and α on n, which matters for independent left and right coefficients.

PRZZ Theorem 4.1 prints the opposite denominator [6, p. 18]. The historical tag `v0.1.0` copied that placement into a general-profile input. The current source removes that axiom and proves the witness input with the density-oriented kernel. For real profiles, exchanging the two finite indices proves $M(P, Q; A, B) = M(Q, P; B, A)$ (`PRZZ.gcdMainTerm_swap`); thus the two main terms coincide for the real witness self-pairing (`PRZZ.witness_densityGcdMainTerm_eq`). This equality holds before limits or differentiation. Both ordered cross terms and the identical operators on the two sides preserve the numerical functional. It is not a justification of the old general mixed-profile statement. The historical tag and its records remain unchanged; a full analytic check of the BCHB original [11] is not claimed here.

5 Arithmetic main terms and the finite functional

The gcd sum in (13) is not defined to be the finite functional below. Their asymptotic identification is a separate part of the proof. The arithmetic calculation retains the unrestricted Dirichlet convolution in (10), extracts its Euler factors and leading poles, and estimates the remainder uniformly on the shift region needed for differentiation.

For $\ell, m \in \{0, 2\}$, the four rational kernels are

$$\begin{aligned}
K_{00}(X, Y, h) &= \frac{XY}{h}, \\
K_{20}(X, Y, h) &= \frac{XY}{h^3} - \frac{2Y}{h^2} + \frac{Y}{Xh}, \\
K_{02}(X, Y, h) &= \frac{XY}{h^3} - \frac{2X}{h^2} + \frac{X}{Yh}, \\
K_{22}(X, Y, h) &= \frac{7XY}{h^5} - \frac{6(X+Y)}{h^4} + \frac{X/Y + Y/X + 8}{h^3} \\
&\quad - \frac{2(X^{-1} + Y^{-1})}{h^2} + \frac{1}{XYh}.
\end{aligned} \tag{16}$$

Use $F_0 = P_1$, $F_2 = P_2$, and define

$$\begin{aligned}
\mathcal{D}_1(A, w; F) &= \theta^{-1}F'(w/\theta) + AF(w/\theta), \\
\mathcal{D}_0(A, w; F) &= F(w/\theta), \\
\mathcal{D}_{-1}(A, w; F) &= \int_0^w e^{-Az}F((w - z)/\theta)\,\mathrm{d}z.
\end{aligned}$$

For each monomial $cX^eY^fh^{-n}$ in $K_{\ell m}$, replace it by

$$\frac{c}{\theta^{\ell+m}}\int_0^\theta \frac{v^{n-1}}{(n-1)!}\mathcal{D}_e(A,\theta-v;F_\ell)\mathcal{D}_f(B,\theta-v;F_m)\,\mathrm{d}v.$$

The resulting function is $H_{\ell m}(A,B)$. Define

$$\mathcal{I}(A,B)=\sum_{\ell,m\in\{0,2\}}\frac{H_{\ell m}(A,B)-e^{-A-B}H_{\ell m}(-B,-A)}{A+B}, \tag{17}$$

$$M_*=q(-\partial_A)q(-\partial_B)\mathcal{I}(A,B)|_{A=B=-R}\,. \tag{18}$$

The dual operation exchanges the two shifts as well as negating them. The evaluation in (18) is away from $A+B=0$.

Proposition 3 (The fixed smooth weighted moment)**.** *Assume Hypotheses 1 and 2. For the fixed data (6)–(11) and every fixed real smooth Φ supported in $[1,2]$,*

$$\lim_{T\to\infty}\frac{1}{T}\int_T^{2T}|U_T(\tfrac12-R/\log T+\mathrm{i}t)\Psi_T(\tfrac12-R/\log T+\mathrm{i}t)|^2\Phi(t/T)\,\mathrm{d}t=M_*\int_1^2\Phi(u)\,\mathrm{d}u. \tag{19}$$

Here is the separation of proof obligations in the implementation. `TwoPieceMainTerm.normalized_witness_uniform` identifies the normalized arithmetic main term uniformly with the finite shift functional. `TwoPieceMoment.normalizedMoment_uniform` combines this with the actual moment estimate. The theorem `TwoPieceMoment.normalizedMoment_exactEvaluator_tendsto` transfers the needed independent left and right shift derivatives through a Cauchy-integral argument. These include all jets through order six on each side, not only derivatives along the diagonal. A fixed bidisc about $(-R,-R)$, with radii sufficiently small, stays away from $A+B=0$. Coefficients are held fixed during these shift derivatives; their height dependence is not differentiated. Finally `WitnessMomentBound.witnessMoment_tendsto` identifies the exact evaluator with the real number in (18) and gives (19).

The Gamma multiplier in the dual approximate functional equation is handled before this identification; it is not replaced by its leading height power by definition. The density-oriented gcd kernel, the two branches, their zero-frequency correction and the remaining Gamma defect are kept distinct until their stated estimates apply. Likewise, defining the inverse operators $\mathcal{D}_e$ does not by itself justify inverse Mellin inversion. The convergence, contour-shift and integration-order lemmas are part of the source chain supplying the arithmetic identification.

6 The numerical gate and its two verification levels

All fixed polynomial coefficients and parameters are rational. The Lean counting proof uses the deliberately conservative bound

$$M_*\le 1.128201<1.128202<\exp\{R(1-0.8845)\}. \tag{20}$$

The first and the strict functional gate are proved in `FixedCertificate`; the positive counting cap $C_*=1.128202$ and its strict comparison with the exponential target are supplied in `WitnessMomentBound`. The interval displayed next is an independent numerical check, not the precision claimed for (20).

Lemma 4 (Independent finite interval calculation)**.** *The interval program accompanying this manuscript encloses (18) and its comparison threshold as follows:*

$$1.12818163807203565629105<M_*<1.12818163807203565629124,$$

$$1.12820541062467513417945<e^{R(1-0.8845)}<1.12820541062467513417947.$$

These statements concern the fixed finite functional, not the analytic moment or the external exponential-sum hypotheses.

The program `paper/numerical_interval.py` uses rational polynomial algebra and interval arithmetic for the exponential pieces. After expansion, polynomial integrals are exact beta integrals. For the truncated exponential in $\mathcal{D}_{-1}$, the normalized j-th shift-derivative remainder is bounded by

$$\frac{3C_F\theta^{1+j}}{j!(1+j)}\frac{(R\theta)^{J+1}}{(J+1)!}.$$

Here j is the derivative order and J is the truncation order; C_F bounds the profile on $[0,1]$, and $R\theta<1$. At truncation order $J=24$, the retained propagated error is less than 8.799×10^{-23}. Runs at precision/order pairs $(40,20)$, $(45,24)$, $(55,24)$, and $(70,24)$ are retained in the certificate, using `mpmath 1.3.0`. This Python arithmetic and its libraries are independently reproducible but are not themselves formally verified in Lean.

The kernel-checked proof of (20) does not read or trust that JSON certificate. The distinction matters: an independent program can give more displayed digits without increasing the logical precision required by the formal theorem. No numerical optimization was performed for this revision.

7 From a weighted moment to actual zero counts

The counting object is not merely the polynomial differential operator U_T. Put $r(u)=q(1/2-u)$; direct rational algebra gives

$$\begin{aligned}r(u)&=0.09491351+0.997010185u+2.089866u^2\\&\quad-2.29671034u^4+1.7701948u^6.\end{aligned}$$

For a fixed height parameter T, define

$$\eta_T(s)=\sum_{j=0}^{6}r_j(\log T)^{-j}\xi^{(j)}(s),\qquad V_T(s)=\eta_T(s)/H(s).$$

The symmetry of ξ makes its even derivatives real and its odd derivatives purely imaginary on the critical line. Since r has only one nonzero odd coefficient, the imaginary part of η_T there is the nonzero real multiple $0.997010185\,(\log T)^{-1}\operatorname{Im}\xi'(s)$. This is the exact xi-prime interface used for argument variation.

Define

$$p_*=1-\frac{\log C_*}{R},\qquad C_*=1.128202.$$

By (20), $p_*>0.8845$. Write $\mathcal{N}_{\xi',c}(T,2T]$ and $\mathcal{N}_{\xi'}(T,2T]$ for the corresponding AF counts in that half-open height window.

Lemma 5 (Eventual dyadic count)**.** *Under Hypotheses 1 and 2, for every real $c<p_*$ there is T_c such that every real $T\ge T_c$ satisfies*

$$c\mathcal{N}_{\xi'}(T,2T]\le\mathcal{N}_{\xi',c}(T,2T]. \tag{21}$$

This is the contract of `WuProportion.dyadic_eventually`. The following points describe how the analytic and geometric conditions are actually supplied.

Exact V_T and the left-line logarithm. The source estimates V_T-U_T in the weighted second moment. It uses nonnegative smooth time weights to pass from (19) to the logarithmic budget, with a strict reserve supplied by $C_*>M_*$. The return to the original left line is proved by reverse Fatou with a dominating positive part and the regular H factor. No unproved endpoint-tightness or counting-gate assumption is used. On the right line the object estimated is the product $V_T\Psi_T=1+O((\log T)^{-1})$; the argument does not assume that the mollifier alone tends uniformly to 1.

Safe inward contours. For fixed $0 < \delta < 1/2$, the horizontal heights are chosen in

$$u \in ((1+\delta)T, (1+\delta)T + 1), \qquad v \in ((2-\delta)T - 1, (2-\delta)T).$$

Actual finite zero sets, boundary nonvanishing and regularity conditions are supplied for these contours. The retained critical-line zeros are included in $(T, 2T]$; discarded endpoint strips are not identified with empty sets. The leading argument-variation budget is

$$\frac{(1-2\delta)T\log T}{2}\left(p_* - \frac{2\varepsilon}{R}\right),$$

with the proved lower-order costs retained in the finite-height inequality. Division by π gives a zero-count budget. One chooses δ and ε using $c < p_*$, before taking T sufficiently large.

The denominator and all terminal heights. The pinned AF Riemann–von Mangoldt result for the actual xi-prime count yields $\mathcal{N}_{\xi'}(T, 2T]/(T\log T) \to 1/(2\pi)$. The local count and this normalization imply (21). AF's additive dyadic-to-cumulative theorem then transfers the inequality to every sufficiently large real terminal height X. Taking $b = (p_* + 0.8845)/2$ proves Theorem 1. This route does not replace the denominator by a zeta count, require simple zeros, or assert a separately supplied arbitrary-window counting adapter.

8 Formal correspondence, reproduction and review

8.1 Principal contracts

All declaration names below lie in the namespace `XiPrime8845`. The repository's generated correspondence table is a navigation aid; where the prose combines several results, it is not a claim that a single theorem has exactly every displayed sentence as its type.

Manuscript item	Lean source contract or verification level
Hypotheses 1 and 2	`PRZZKloostermanInputs.WeilIncomplete` and `PRZZKloostermanInputs.DeshouillersIwaniecLemma1`; explicit assumptions, not proved inputs.
Proposition 2	`PRZZFamilyMomentOptimal.moment_bound`, with a fixed admissible partition seed and $K \geq 4$.
Public witness estimate	`PRZZWitnessBridge.witness_estimate` and `PRZZ.przz41_case3_K2_witness`; the latter is now a theorem in `PRZZProvedInput.lean`.
Proposition 3	`WitnessMomentBound.witnessMoment_tendsto`, through `TwoPieceMainTerm` and `TwoPieceMoment`.
Conservative gate (20)	`FixedCertificate.actual_upper` and `WitnessMomentBound.countingCap_lt_target`.
Lemma 4	Independent Python interval certificate; its narrow decimal endpoints are not asserted as a matching Lean theorem.
Lemma 5	`WuProportion.dyadic_eventually`, with the actual AF counts.
Theorem 1	`WuProportion.strict_proportion` and `WuProportion.ratio_eventually_gt`.

8.2 Pinned sources and the scope of the recorded checks

The mathematical baseline for this editorial revision is commit `2418fd5d334dbd7b4d99baaba5d8c9f41681884a`. The toolchain is Lean `v4.33.0-rc2`, with AF commit `3635e74826a4c1fcece7d1cd2b6fa75e43a00510` and Mathlib commit `51e6992efd06126df61a496bebf8f49482a4e129`. These fixed inputs, rather than moving upstream branches, determine the verification environment. The public proof/software snapshot is release `v0.2.0` [8], at commit `8fbfbe376d879274c5be7e2f655551c967dfdf99`, archived as doi:10.5281/zenodo.24101490. The entire `lean/` source tree is byte-identical at the mathematical baseline and the release commit; the intervening changes concern documentation, verification records and supporting verification tools. The present bibliographic and submission-metadata revision postdates that frozen archive; it does not replace the archived manuscript bytes.

The accepted record `checks/clean-verification-2026-11-10-local.json` covers 778 local source modules and 4,405 registered printed axiom audits. The headline import closure contains 730 local modules. All 778 modules were freshly compiled in an initially empty output directory, with zero reused local records. The linked upstream record `checks/clean-verification-2026-11-10-af.json` reports fresh compilation of all 101 required pinned AF modules, with no old AF outputs. Only fixed Mathlib/package caches were reused; unrelated AF targets were not rebuilt. Compile times were 190.253 seconds (AF) and 2,939.546 seconds (local, six worker slots). Older records, including the one-fresh/777-reused invocation, are preserved unchanged as history.

The four headline audits (the four theorems audited at the end of `WuProportion`) list exactly

$$\{\texttt{propext},\ \texttt{Classical.choice},\ \texttt{Quot.sound}\}.$$

Neither an old PRZZ axiom nor `sorryAx` appears in those audits. However, an axiom list does not list the assumptions supplied as ordinary theorem arguments: the two explicit exponential-sum hypotheses remain indispensable. Kernel checking proves the conditional implication, not the cited analytic inputs.

The reviewer entry point is `docs/REVIEW_GUIDE.md`. It links the dependency graph, exact trust base, module table, source-to-paper correspondence and reproducibility instructions. The Python release verifier checks distribution hashes, links, and the independent finite certificate. Those checks are distinct from Lean compilation and from the visual inspection of this PDF. The manuscript/PDF revision changes neither the mathematical Lean sources nor the numerical algorithm or fixed witness.

8.3 Questions requiring independent analytic review

The principal external review is the exact match between Hypotheses 1–2 and their cited sources: normalization, inverse phases, coprimality restrictions, coefficient dependence and uniform constants all matter. A second review layer is the correspondence of the source definitions with the intended xi-prime problem: density-oriented shifts, the unrestricted Feng-type convolution [9], Gamma normalization, height-dependent basis uniformity, and the actual multiplicity counts. Finally, the finite main-term functional and its independent numerical implementation should be compared separately from the moment and counting arguments.

These are specification and mathematical-review questions, not hidden hypotheses added to the Lean theorem. Conversely, a successful compiler run must not be reported as their independent resolution. At the time of writing, no independent referee report has been received. No simple-zero improvement or priority claim is made.

Data and code availability

The Lean sources, Python finite-certificate code, saved numerical outputs, dependency pins, verification records and reproduction instructions are publicly available in the repository https://github.com/miosync-masa/xiprime-two-piece-mollifier. The immutable software snapshot cited here is `v0.2.0`, commit `8fbfbe376d879274c5be7e2f655551c967dfdf99`, released on 10 November 2026 and archived on Zenodo [8], doi:10.5281/zenodo.24101490. Start with `docs/REVIEW_GUIDE.md` and `docs/REPRODUCIBILITY.md` at that tag, not a moving branch. The two clean-build records cited in Section 8 are included in the release. No empirical dataset or zero table is used; the reusable data are the exact witness, interval outputs and compiler audit records. Mathlib and AF are external dependencies fixed to the commits above. This revised manuscript is not claimed byte-identical to the PDF in the frozen `v0.2.0` archive.

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Declaration of competing interests

The author declares no competing interests.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

The author used Claude (Anthropic) and ChatGPT/Codex (OpenAI) during exploratory mathematical discussions, Lean and Python code development and debugging, literature lookups, and the organization and revision of the manuscript. The author directed the work, reviewed the outputs and revised the included material, and accepts responsibility for the content. AI tools are not authors. Kernel checking and reproducible finite calculations do not replace that responsibility or independent review of the analytic specifications.

References

[1] A. Weil, On some exponential sums, *Proc. Natl. Acad. Sci. USA* **34** (1948), 204–207. doi:10.1073/pnas.34.5.204.

[2] G. Harcos, *Weil's bound for Kloosterman sums*, expository notes, Theorem 1. https://users.renyi.hu/~gharcos/weil.pdf.

[3] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, American Mathematical Society Colloquium Publications, vol. 53, 2004, Corollary 11.12. Author-hosted chapter excerpt: https://people.math.ethz.ch/~kowalski/ik-ant-exp-sums.pdf.

[4] J.-M. Deshouillers and H. Iwaniec, Power mean-values for Dirichlet's polynomials and the Riemann zeta-function, II, *Acta Arith.* **43** (1984), 305–312, Lemma 1. doi:10.4064/aa-43-3-305-312.

[5] X. Wu, Distinct zeros of the Riemann zeta-function, *Q. J. Math.* **66** (2015), no. 2, 759–771. doi:10.1093/qmath/hav014. Preprint: arXiv:1206.3737v2; the argument used here was inspected in Section 3 of that version.

[6] K. Pratt, N. Robles, A. Zaharescu and D. Zeindler, *More than five-twelfths of the zeros of ζ are on the critical line*, *Res. Math. Sci.* **7** (2020), Article 2. doi:10.1007/s40687-019-0199-8. The theorem and page references used here refer to arXiv:1802.10521v3, Theorem 4.1 and its proof (printed p. 18 onward).

[7] L. Alpöge and R. Furman, *More than two thirds of the zeta zeros are simple and on the critical line*, arXiv:2608.13637v2; associated *formal-math* Lean library, tag `v1.0`, pinned commit specified in Section 8. https://github.com/anthropics/formal-math.

[8] M. Iizumi, *miosync-masa/xiprime-two-piece-mollifier: v0.2.0 — The PRZZ axiom is gone: 88.45% from explicit Weil and Deshouillers–Iwaniec hypotheses*, software release, 10 November 2026. doi:10.5281/zenodo.24101490. Commit `8fbfbe376d879274c5be7e2f655551c967dfdf99`. https://github.com/miosync-masa/xiprime-two-piece-mollifier/releases/tag/v0.2.0.

[9] S. Feng, Zeros of the Riemann zeta function on the critical line, *J. Number Theory* **132** (2012), no. 4, 511–542. doi:10.1016/j.jnt.2011.10.002.

[10] J. B. Conrey, More than two fifths of the zeros of the Riemann zeta function are on the critical line, *J. reine angew. Math.* **399** (1989), 1–26. doi:10.1515/crll.1989.399.1.

[11] R. Balasubramanian, J. B. Conrey and D. R. Heath-Brown, Asymptotic mean square of the product of the Riemann zeta-function and a Dirichlet polynomial, *J. reine angew. Math.* **357** (1985), 161–181. doi:10.1515/crll.1985.357.161.

[12] D. R. Heath-Brown, Prime numbers in short intervals and a generalized Vaughan identity, *Canad. J. Math.* **34** (1982), no. 6, 1365–1377. doi:10.4153/CJM-1982-095-9.

[13] N. Levinson, More than one third of zeros of Riemann's zeta-function are on $\sigma = 1/2$, *Adv. Math.* **13** (1974), no. 4, 383–436. doi:10.1016/0001-8708(74)90074-7.

[14] J. B. Conrey, Zeros of derivatives of Riemann's xi-function on the critical line, *J. Number Theory* **16** (1983), no. 1, 49–74. doi:10.1016/0022-314X(83)90031-8.

[15] L. de Moura and S. Ullrich, The Lean 4 theorem prover and programming language, in *Automated Deduction — CADE 28*, Lecture Notes in Computer Science **12699**, Springer, 2021, 625–635. doi:10.1007/978-3-030-79876-5_37.

[16] The mathlib Community, The Lean mathematical library, in *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, ACM, 2020, 367–381. doi:10.1145/3372885.3373824.